

Why Behaviour-oriented Concurrency?

Inspired by actors but decouples isolation and concurrency



Concurrent behaviours with ad-hoc state access

The state of BoC today

When Concurrency Matters: Behaviour-Oriented Concurrency

LUKE CHEESEMAN, Imperial College London, UK
MATTHEW J. PARKINSON, Microsoft Azure Research, UK
SYLVAN CLEBSCH, Microsoft Azure Research, UK
MARIOS KOGIAS, Imperial College London, UK Microsoft Research, UK
SOPHIA DROSSOPOULOU, Imperial College London, UK
DAVID CHISNALL, Microsoft, UK
TOBIAS WRIGSTAD, Uppsala University, Sweden
PAUL LIÉTAR, Imperial College London, UK

$$\begin{array}{l} \text{STEP} \frac{E, h \hookrightarrow E', h'}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R \uplus (\bar{\kappa}, E'), P, h'} \quad \text{SPAWN} \frac{E \hookrightarrow_{\text{when } (\bar{\kappa}') \{E''\}} E'}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R \uplus (\bar{\kappa}, E'), P : (\bar{\kappa}', E''), h} \\ \\ \text{RUN} \frac{(\bigcup_{(\bar{\kappa}', _) \in (P' \cup R)} \bar{\kappa}') \cap \bar{\kappa} = \emptyset}{R, P' : (\bar{\kappa}, E) : P'', h \rightsquigarrow R \uplus (\bar{\kappa}, E), P' : P'', h} \quad \text{END} \frac{\text{finished}(E)}{R \uplus (\bar{\kappa}, E), P, h \rightsquigarrow R, P, h} \end{array}$$